

RF Sim for KiCad

Installation Guide

From the download page to the first simulation

This guide is for someone who has never seen RF Sim. It follows the way you will go: download, install, switch on KiCad's plugin interface, start KiCad, let RF Sim fetch its solver, and run a first test. Most of the time goes into waiting for downloads.

Pre-release – please read. RF Sim is a student project and not a finished product. Results can contain errors; check important designs with a measurement or a professional tool. Simulations take long: minutes for a first look, hours for final numbers on an older PC (openEMS runs on the CPU only). The time estimate is cautious; the real run usually ends sooner. Shorter run times are the main work in progress.

You need	
Windows 10 or 11	RF Sim was tested on Windows only.
KiCad 10	The plugin interface of older KiCad versions is not supported.
Internet access	For the downloads below. Afterwards RF Sim works offline.
Free disk space	For openEMS (about 65 MB download) and KiCad's Python packages.

1 Download

1. Open the project's release page on GitHub: `github.com/at-kat-454/kicad-rfsim-releases` → **Releases**.
2. Open the newest release and unfold **Assets**.
3. Download `kicad-rfsim-pcm-<version>.zip`. Do *not* unzip it.

The other files are optional:

File	Content
`kicad-rfsim-pcm-<version>.zip`	**the plugin**, for KiCad's Plugin and Content Manager
`kicad-rfsim-plugin-<version>.zip`	the same plugin for a manual install
`kicad-rfsim-docs-<version>.zip`	installation guide, user manual, worked example, verification report (PDF and LaTeX)
`kicad-rfsim-boards-<version>.zip`	test boards for your own validation: KiCad projects with ready setups and reference values, no simulation results
`SHA256SUMS.txt`	checksums of all files

To check a download, open a Command Prompt in the download folder and run `certutil -hashfile kicad-rfsim-pcm-<version>.zip SHA256`. The result must equal the line of that file in `SHA256SUMS.txt`.

2 Install the plugin

Choose one of the two ways. The first is the quickest, the second also delivers updates.

2.1 From the downloaded file

1. Start KiCad and open **Plugin and Content Manager** in the main window.
2. Press **Install from File...** and select `kicad-rfsim-pcm-<version>.zip`.
3. Press **Apply Pending Changes**. Wait until the manager reports success.

2.2 From the repository, with updates

1. In the **Plugin and Content Manager** press **Manage...**.
2. Add the repository `https://at-kat-454.github.io/kicad-rfsim-releases/repository.json` and press **Save**. Wait until the repository is shown as loaded.
3. Open the tab **Plugins**, choose the repository *RF Sim for KiCad*, select *RF Sim* and press **Install**.
4. Press **Apply Pending Changes**.

New versions then show up as updates in the same list.

> Install RF Sim only once. A copy from an earlier manual install in `Documents\KiCad\10.0\plugins` and the managed copy in `Documents\KiCad\10.0\3rdparty\plugins` exist side by side, and KiCad may load the old one. Delete the old folder `io.github.kicad-rfsim` first.

3 Switch on the KiCad API

RF Sim talks to KiCad through KiCad's plugin interface. Check that it is switched on.

1. In the KiCad main window open **Preferences** → **Preferences...** → **Plugins**.
2. Tick **Enable KiCad API**.
3. Look at the field **Python interpreter**. It must point to KiCad's own interpreter, `C:\Program Files\KiCad\10.0\bin\pythonw.exe`. Below it KiCad shows the Python version it found. If the field is empty or wrong, press **Auto Detect**.
4. Press **OK** and **close all KiCad windows**.

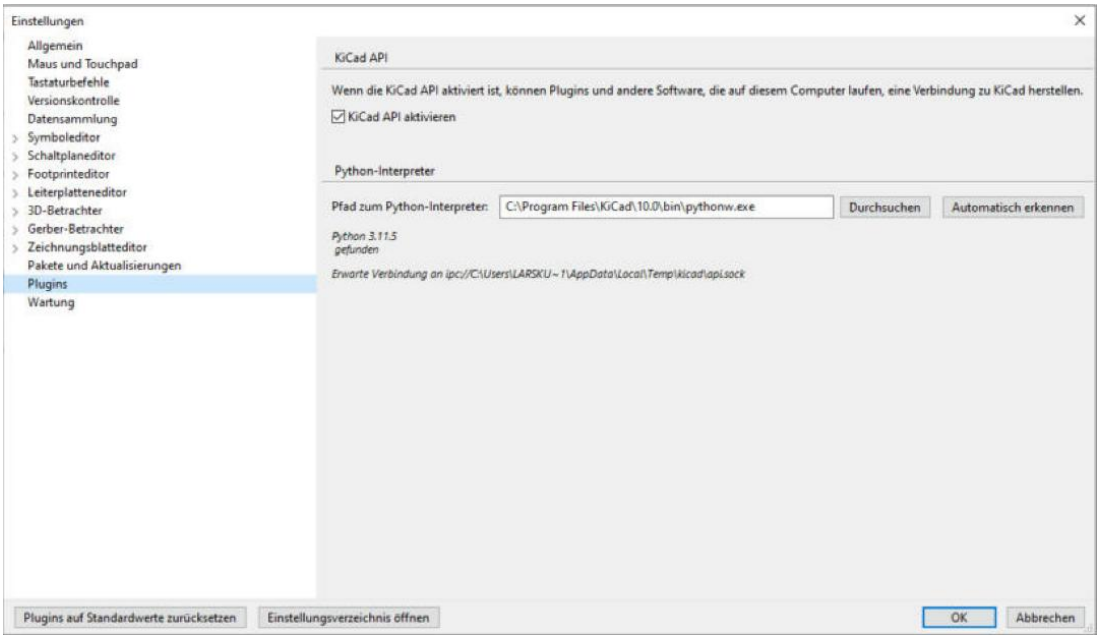


Figure 1: Preferences, page Plugins: API enabled, interpreter is KiCad's `pythonw.exe` (German user interface).

4 First start of KiCad

1. Start KiCad and open a board in the PCB editor.
2. KiCad now installs the Python packages that RF Sim needs (numpy, scipy, shapely, matplotlib, scikit-rf, plotly, gmsh, h5py, wxPython, psutil and kicad-python). This needs the internet and takes a few minutes. Keep KiCad open until it is done.
3. The **RF Sim** button appears in the toolbar of the PCB editor. If it does not, restart KiCad once more.

You can have several KiCad instances open. RF Sim opens one window per board, so a running simulation on one board does not block you from looking at the results of another.

5 Get the solver (openEMS)

The field solver openEMS is licensed under the GPL and therefore not part of the plugin. RF Sim downloads it.

1. Press **RF Sim**, go to the page **Simulate** and start a simulation.
2. RF Sim offers to download the official Windows build of openEMS 0.37 (about 65 MB). Accept.
3. openEMS is stored in `%LOCALAPPDATA%\kicad-rfsim\openEMS`.

openEMS already on the PC, or the download is blocked (company network, proxy, virus scanner): copy the openEMS folder to the PC. In RF Sim open **Solver Settings**, press **Browse**, select the folder that contains `openEMS.exe`, then press **Search Again**. The page shows the path and the version when it works.

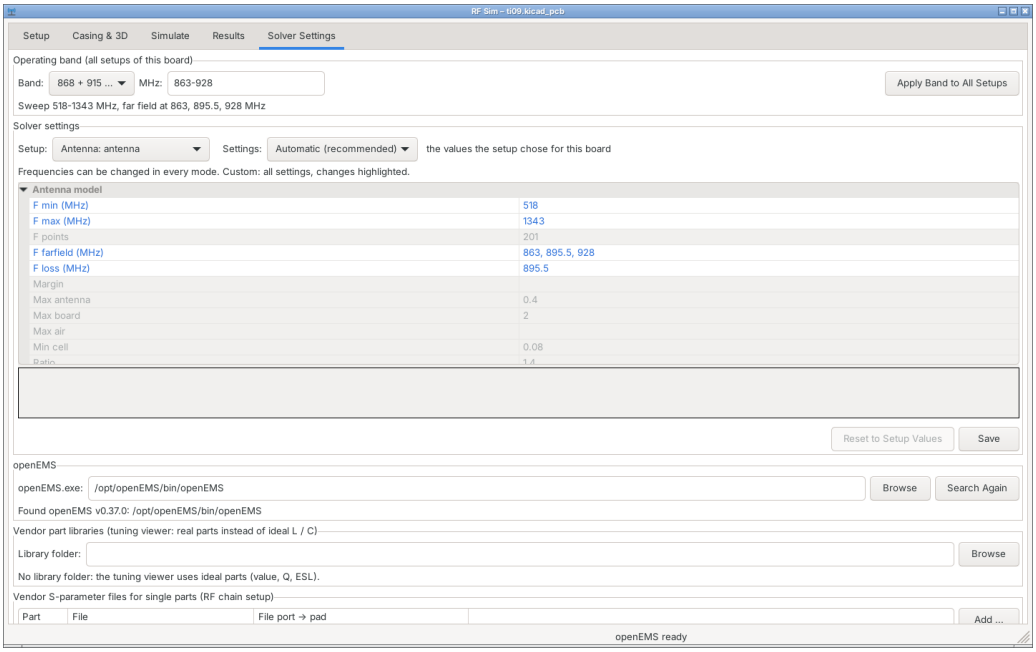


Figure 2: RF Sim page **Solver Settings**: the group **openEMS** shows path and version.

6 First test

Run a small known case before your own board.

1. Unzip `kicad-rfsim-boards-<version>.zip`.
2. Open the project `validation\ms50\ms50.kicad_pro` in KiCad and open its PCB. This is a 50 Ω microstrip line with a ready setup.
3. Press **RF Sim** and go to **Simulate**. Tick the setup and press **Start Simulation**.
4. When the run is finished, open **Results** and compare the line impedance with the expected value of about 50 Ω (±2 Ω).

`validation\ms50\README.txt` lists the reference values. If the numbers fit, the installation is complete. Continue with the *Worked Example* or the user manual, section *Quick start*.

7 Problems

Symptom	What to do
The **RF Sim** button is missing	Check **Enable KiCad API** and the interpreter (section 3). Restart KiCad. Check that the plugin is listed under **Installed** in the Plugin and Content Manager.
The packages never finish installing	The first start needs the internet. Behind a proxy or firewall the package installation fails: connect without it once.
The old version starts	Two copies of the plugin exist. Delete the old folder (section 2).
"openEMS not found" or "did not start"	**Solver Settings** → **Browse** or **Search Again**. Let RF Sim download it, or copy a working openEMS folder. A virus scanner may block `openEMS.exe`: allow it.
RF Sim window does not open a second board	One window exists per board. A second press raises the existing window. Open the other board in its own KiCad instance.
Plugin Manager cannot add the repository	Open the `repository.json` address in a browser. If it does not show text, the network blocks it. Use the file install.

8 Update and uninstall

- **Update:** install the new zip the same way, or press the update in the Plugin and Content Manager. Results of older versions stay usable.
- **Uninstall:** Plugin and Content Manager → **Installed** → *RF Sim* → **Uninstall** → **Apply Pending Changes**.
- **Remove the data:** delete `%LOCALAPPDATA%\kicad-rfsim` (openEMS, model cache, board snapshots). Your boards and the `rfsim` folders inside your projects stay.